

Marena Abboud

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EDUCATION

Moravian University, Bethlehem, PA

May 2026

B.S. Computer Science, Minor in Art · GPA 3.88, Summa Cum Laude

PROJECT EXPERIENCE

MindScape 360, VR Mindfulness App | SOAR Research Fellowship

Summer 2025

github.com/MoravianUniversity/Mindscape-360

- Designed and published a cross-platform VR mindfulness app for acute-care patients, built to minimize onboarding steps so users in pain can start sessions right away.
- Partnered with a Doctor of Physical Therapy to align the UI with clinical research on accessibility and stress reduction. Built in Kotlin and Jetpack Compose with C++, integrating MediaPlayer on Android and AVPlayer on iOS for synchronized guided audio and visuals.
- Published on the App Store and Google Play; currently piloting at Good Shepherd Rehabilitation Hospital. Named a URSCA All-Star by Moravian's Rehabilitation Sciences Occupational Therapy program.

Mirrulations Search, Federal Regulations Search Platform

Fall 2025 – Spring 2026

github.com/mirrulations/mirrulations-search

- Worked directly with clients from CareSet and Merck in an agile environment to build a search platform helping researchers and policy professionals navigate federal rulemaking across dockets, documents, and comments from regulations.gov.
- Designed and implemented the full document download system from scratch: REST API endpoints, a Redis-backed async worker packaging and uploading files to S3, a background cleanup job, PostgreSQL schema changes, EC2 deployment with systemd, and frontend integration.
- Also implemented Google OAuth authentication, CFR part filtering, and sort-by functionality across date, comment count, and document count. Stack: Python, Flask, PostgreSQL, OpenSearch, Redis, AWS S3/EC2, React.

BytePath, Full-Stack Python Learning Platform | Moravian University

Fall 2025 – Present

- Built a production Python learning platform used by ~30 students across two intro CS courses, featuring leveled question sets, per-student progress tracking, and instructor dashboards.
- Built a Flask and SQLAlchemy REST API with 6 route blueprints, paired with a React and TypeScript frontend; integrated Google OAuth and deployed to AWS EC2 with automated systemd services.
- Kept building after the semester ended, adding CSV roster management with soft-delete and audit history, per-question analytics, and a realistic seeded dataset for instructor testing.

Collegiate Hackathon | Lehigh University

Fall 2025

- Built a web app converting notes and audio into flashcards and quizzes on a serverless AWS architecture (API Gateway, Lambda, Bedrock, DynamoDB) with a React + Tailwind + Vite frontend.

Phishing-URL Classifier | Neural Networks Course Project

Spring 2024

- Developed a phishing-URL classifier (~95% accuracy) using Word2Vec embeddings and a Ridge Classifier, with tokenization and feature engineering in pandas, scikit-learn, and gensim.

LEADERSHIP & TEACHING

Teaching Assistant & PASS Leader, CS1 (Python) · Moravian University

Fall 2024 – Spring 2026

Tutor, Quantitative Resource Center (Python & Calculus I) · Moravian University

2023 – 2024

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Tools: Flask, React, Jetpack Compose, PostgreSQL, Redis, OpenSearch, AWS (EC2, S3, Lambda, API Gateway, DynamoDB, Bedrock), Docker, Tailwind

WORK EXPERIENCE

Server / Hostess · Mama Nina's Italian Restaurant (2023 – Present) · Hometown Family Restaurant (2020 – 2023)

Balanced customer-facing work with a full course load all four years of college.